

Predictive Saturation Management through a Behavior--Realization Interface

Anonymous submission

Abstract

Clipping protects actuator bounds but can silently destroy a fast controller's designed or trained closed-loop behavior. We present a two-rate predictive realization architecture anticipating this loss without replacing the controller. A proportional--derivative, impedance, reinforcement-learning, neural-network, or behavior-conditioned controller runs unchanged at 1 kHz; a 50 Hz manager keeps its request inside a robot-specific, uncertainty-tightened feasible set evaluated along the nominal rollout; a logged directional-authority indicator exposes loss hidden by scalar utilization. The main result gives sufficient conditions for a behavior-coordinate certificate to transfer to a physical robot: realization uncertainty must fit inside the actuator margin, and the successor defect inside the certificate margin. A deterministic study of 108 configurations spans five controllers, three realization maps, eight scenarios, and six ablations. Full-horizon enforcement removes a 3.594 Nm violation missed by a first-step constraint, and a cross-realization audit finds all successor defects below 0.00759 m/s against a common 0.03 m/s threshold. The method prevents sampled violations under slow saturation, directional-authority collapse, and near-boundary braking; abrupt disturbances and severe model or preview mismatch exceed the audit conditions, marking the result's boundary. The study supports conditional certificate transfer and predictive saturation management in reduced-order simulation; it does not experimentally instantiate the abstract certificate, establish universal policy safety, or demonstrate real-time hardware performance.

Index Terms— physical interaction, actuator saturation, predictive constraint management, control refinement, model predictive control.

I. Introduction

Robot control software increasingly combines high-rate feedback with lower-rate prediction. The fast layer may be a conventional proportional--derivative (PD) controller, an impedance controller, or a

learned policy. Its purpose is to react to sensor feedback with low latency. A slower layer can reason over a horizon, but embedding every nominal controller inside a new robot-specific model-predictive controller sacrifices the modularity that makes the fast layer useful.

These fast controllers can approach saturation through different channels. Before command limiting, a fixed-gain PD law $\tau \propto k_p(q_{\text{goal}} - q) - k_d\dot{q}$ maps a sufficiently large tracking error to a large actuator request. An impedance law introduces interaction force through $M_d\ddot{e} = -K_d e - D_d\dot{e} + F_h$, so contact can increase the requested acceleration even when tracking error is modest. The implementation studied here imposes a common acceleration bound on every nominal interface, but this does not guarantee torque feasibility after configuration-dependent realization.

The same distinction applies to bounded learned policies. The $\tanh$ -squashed policy used in Section VI bounds its requested acceleration, but the required torque is $\tau = \tau_{\text{base}}(x) + H_r(x)a_{\text{req}}$; a fixed bound on a_{req} does not imply a configuration-independent bound on τ . A neural policy fitted to impedance demonstrations can reproduce the same force-dependent requests without exposing interpretable stiffness and damping parameters. An upstream learned or language-conditioned module may likewise propose a motion primitive without representing the actuator geometry of the executing robot. These observations motivate a common realization interface, not a claim that every learned behavior is certifiable.

The shared consequence is an acceleration request that the current robot cannot realize in its current configuration. We therefore treat saturation management as an interface problem. Controller-specific measures remain useful, but the robot-aware realization layer developed here anticipates the common physical failure without requiring its predictive optimization to be reformulated for each upstream controller.

Formally, let a nominal controller of any of these kinds request

$$\tau_k^0 = \pi_\theta(s_k),$$

where s_k is the controller observation and π_θ may be analytic or learned. A conventional implementation applies

$$\tau_k^{\text{app}} = \text{proj}_{\mathcal{T}_r}(\tau_k^0), \quad \mathcal{T}_r = \{\tau : \tau_{\min,r} \leq \tau \leq \tau_{\max,r}\}.$$

Projection protects the actuator command but changes the acceleration generated by the robot. After projection, the robot is no longer realizing the nominal PD, impedance, or learned closed-loop dynamics. This distinction is especially consequential in physical interaction control, where the requested acceleration determines apparent compliance and disturbance response.

The architecture developed below requires more than an arbitrary τ_k^0 : it requires the nominal controller to expose the task-acceleration request that τ_k^0 realizes, so that the manager's correction is well defined (Section III). All five controller interfaces evaluated in Section VI already expose this request directly.

We study whether prediction can preserve the nominal fast controller while correcting its requested dynamics before the actuator command becomes unrealizable. The proposed architecture separates two responsibilities. A behavior layer specifies the nominal local dynamics at 1 kHz. A predictive realization manager, executing at 50 Hz, checks whether those dynamics can be produced by the current robot over a finite horizon and computes a minimum correction when required. A final high-rate projection remains necessary for disturbances that arrive between predictive updates.

Feedback linearization to a double-integrator-like behavior model is classical [1], [2]. Likewise, reference governors, predictive safety filters, anti-windup control, and model-predictive constraint handling are established ideas [3]--[7]. The contribution is therefore not another claim of universal prediction. The central question is instead:

When can a behavior-level certificate be reused across different nominal controllers and different robots, despite configuration-dependent actuator saturation?

The answer requires preserving, rather than hiding, the robot-specific realization geometry. The common behavior dynamics provide a reusable location for prediction and certification, while the actuator-feasible acceleration set and uncertainty bounds remain specific to each robot.

The contributions are:

1. a two-rate architecture that retains an existing 1 kHz controller and uses slower MPC only to anticipate and correct loss of actuator realizability;
2. a robot-dependent acceleration-zonotope representation and a directional-authority indicator logged at every predictive update, exposing failures hidden by scalar torque utilization;
3. a conditional certificate-transfer theorem that separates a reusable abstract certificate from robot-specific actuator and successor-error tests; and
4. a reproducible 108-configuration simulation study evaluating controller substitution, a sampled cross-realization interface audit, horizon-wide constraints, uncertainty tightening, final high-rate projection, and failure outside the tested operating region.

The experiments are reduced-order and intentionally include negative cases. They establish the mechanism and its logical limits rather than hardware-level universality.

II. Related Work

Impedance control specifies a desired dynamic relation between motion and interaction force [1], while operational-space control maps task accelerations or wrenches to joint torque [2]. Both rely on sufficient actuator authority. Direct clipping preserves torque bounds but changes the realized dynamics. Anti-windup methods explicitly address saturation-induced performance and stability degradation [6], [7], but are commonly designed around a particular closed-loop controller and often react to saturation rather than forecasting a configuration-dependent loss of task authority.

Model-reference adaptive impedance control instead asks the robot to reproduce a prescribed impedance model despite uncertain physical dynamics [11]. This is close to our behavior-coordinate viewpoint, but it does not by itself provide finite-horizon actuator-constraint management. The present work retains the fast interaction controller and alters its requested task acceleration before hard clipping is predicted. The final projection is not removed; it remains a last-resort actuator guard.

Reference and command governors modify a reference supplied to a pre-stabilized system so that predicted constraints remain satisfied [3]. Predictive safety filters similarly modify nominal actions using a predictive model and a recoverable terminal condition [5]. Robot-specific MPC can incorporate full nonlinear dynamics and actuator constraints directly. These approaches establish that predictive constraint management is valuable; the present work does not claim otherwise.

Recent interaction-control work combines MPC with impedance adaptation in several ways. Roveda *et al.* optimize impedance setpoints and damping using learned interaction dynamics [12]; Haninger *et al.* jointly optimize trajectory and impedance using Gaussian-process task models [13]; Anand *et al.* use a learned Cartesian impedance model inside MPC to adapt variable-impedance parameters across manipulation tasks [14]; and Xue *et al.* combine model-predictive variable impedance control with environment estimation, passivity, and safety-oriented mode switching [15]. In these methods, impedance parameters or trajectories are optimization variables. Our manager instead treats the upstream controller as fixed and selects a physically realizable task-acceleration sequence close to its request.

This difference does not make the proposed optimizer categorically separate from a reference governor or predictive safety filter. Either architecture could implement the realization manager. The contribution lies in the behavior--realization contract and the sufficient transfer condition, not in assigning a new name to constrained MPC. Unlike predictive safety filters with a verified terminal safe set [16], the present simulation enforces finite-horizon state and actuator constraints but does not establish recursive feasibility.

Control barrier functions provide a natural high-rate mechanism for enforcing state inequalities [4]. They are useful as the final projection in the proposed architecture, but an instantaneous constraint may intervene only after the state has approached a region from which actuator authority is insufficient. Our predictive layer instead evaluates future realization geometry and finite-horizon running constraints; the present implementation does not use a terminal recoverability condition.

The theoretical lineage is approximate simulation and control refinement [8], [9]. An interface maps an abstract input to a concrete input while bounding the mismatch between abstract and concrete successors. Contract-based design similarly separates assumptions about a component from the guarantees it delivers [10]. We specialize that logic to actuator-limited robot realization: the refinement defect is constructed from a tightened actuator set, a torque-realization error bound, and an abstract successor-error bound. This makes certificate transfer conditional and falsifiable.

III. Problem Formulation

Consider a torque-controlled robot r ,

$$M_r(q)\ddot{q} + h_r(q, \dot{q}) = \tau + J_r(q)^\top F_h,$$

where $M_r(q) \succ 0$, h_r collects modeled bias terms, F_h is an interaction wrench, and $\tau \in \mathcal{T}_r$.

The actuator set is the box

$$\mathcal{T}_r = \{\tau : \tau_{\min, r} \leq \tau \leq \tau_{\max, r}\},$$

with both inequalities interpreted componentwise.

The architecture requires the fast controller to expose a nominal task-acceleration request, not an arbitrary torque:

$$\mathcal{I}_\theta(s_k, \xi_k) \mapsto (a_{\text{req}, k}^0, \mathcal{P}_{\theta, k}, \Sigma_{\theta, k}).$$

Here $a_{\text{req}, k}^0$ is the current nominal task-acceleration request, $\mathcal{P}_{\theta, k}$ is an optional preview operator, and $\Sigma_{\theta, k}$ bounds preview uncertainty. An analytic PD or impedance controller can be evaluated along predicted states. A learned policy may be queried on predicted observations. If no validated preview is available, zero-order hold or a learned predictor may be used, but its error must enter $\Sigma_{\theta, k}$. A black-box controller with neither query access nor bounded preview error is outside the certificate.

The nominal pre-projection torque is then defined, not assumed, as the robot's own realization of that request (the realization map is given below),

$$\tau_k^0 = \tau_{\text{base},r}(x_k) + H_r(x_k)a_{\text{req},k}^0,$$

so the fast-loop correction of Section IV is well defined by construction. A controller that instead outputs torque directly is outside this interface unless it exposes the decomposition $\tau_k^0 = \tau_{\text{base},r}(x_k) + H_r(x_k)a_{\text{req},k}^0 + \tau_{\perp,k}^0$, with the residual $\tau_{\perp,k}^0$ folded into the actuator-budget accounting of $\tau_{\text{base},r}$ below; otherwise its realizability cannot be tested against $\mathcal{A}_r^{\text{tight}}(x)$. All five controller interfaces evaluated in Section VI already expose a_{req}^0 directly and require no such decomposition.

Let

$$z = [e^\top \quad \dot{e}^\top]^\top$$

denote a task-error state. Over a local operating region, the requested behavior can be written

$$\ddot{e} = a_{\text{req}} + d_r,$$

where $a_{\text{req}} \in \mathbb{R}^d$ is the requested task acceleration and d_r contains matched interaction and modeling effects. Unmatched discretization, coupling, and force errors will be retained later as a set-valued successor defect.

At the manager period Δt , the abstract predictor is

$$z_{\ell+1} = Az_\ell + B(a_{\text{req},\ell} + d_\ell),$$

$$A = \begin{bmatrix} I & \Delta t I \\ 0 & I \end{bmatrix}, \quad B = \begin{bmatrix} \frac{1}{2}(\Delta t)^2 I \\ \Delta t I \end{bmatrix}.$$

This model is a common interface. Let $\tau_{\text{base},r}(x)$ include bias compensation and every secondary command that consumes actuator authority, including orientation and null-space torque. A local realization map is

$$\tau = \tau_{\text{base},r}(x) + H_r(x)a_{\text{req}},$$

with, for example,

$$H_r(x) = J_r(q)^\top \Lambda_r(q), \quad \Lambda_r(q) = (J_r M_r^{-1} J_r^\top)^{-1}.$$

Because the robot dynamics above already include $J_r(q)^\top F_h$, the actuator only supplies the part of the task force not already delivered by the interaction wrench; $\tau_{\text{base},r}(x)$ and the realization map are evaluated net of this feedforward term. Consequently $\hat{\tau}_r$ and τ_r^{pre} , introduced in Section IV, depend

on the current or forecast F_h as well as on (x, a_{req}) ; we suppress this argument notationally rather than write it at every occurrence.

The acceleration request is realizable only if

$$a_{\text{req}} \in \mathcal{A}_r(x) = \{a_{\text{req}} : \tau_{\min,r} \leq \tau_{\text{base},r}(x) + H_r(x)a_{\text{req}} \leq \tau_{\max,r}\}.$$

Equivalently, if local task acceleration is $a = b_r(x) + G_r(x)\tau$, the actuator box maps to the zonotope

$$\mathcal{Z}_r(x) = b_r(x) + G_r(x)\tau_c + G_r(x) \text{diag}(\Delta\tau)[-1, 1]^{n_r},$$

where

$$\tau_c = \frac{\tau_{\max,r} + \tau_{\min,r}}{2}, \quad \Delta\tau = \frac{\tau_{\max,r} - \tau_{\min,r}}{2}.$$

For a full-dimensional d -dimensional zonotope with n_r generators in general position, every choice of $d - 1$ generators determines a facet normal and a pair of opposing supporting facets. Hence the number of facets is $2 \binom{n_r}{d-1}$ (and no larger in degenerate cases). Its support function in direction y is the support of its center plus the sum of the absolute projected generators, so directional authority can be evaluated without enumerating all 2^{n_r} box vertices.

Thus the behavior dynamics can share coordinates, but their feasible set cannot be robot-independent.

Given an arbitrary nominal controller that supplies a_{req}^0 or an equivalent torque command, determine a correction Δa_{req} such that:

1. $a_{\text{req}} = a_{\text{req}}^0 + \Delta a_{\text{req}}$ remains close to the nominal requested behavior;
2. the uncertain physical realization remains inside $\mathcal{T}_r$ over the prediction horizon;
3. the predicted state satisfies the running workspace and speed constraints; and
4. the robot-specific realization can be tested against sufficient conditions for transferring an independently established abstract certificate.

IV. Predictive Realization Manager

The nominal controller, state-dependent realization map, command-rate limiter, and final actuator projection execute every $T_f = 1$ ms. The predictive manager executes every $\Delta t = 20$ ms. Let $a_{\text{req},i|\ell}$ be the complete acceleration request, rather than a correction variable. The implemented

manager rolls out the nominal controller, evaluates the realization map along that nominal rollout, and solves

$$\begin{aligned}
& \min_{a_{\text{req},0:N-1}} \sum_{i=0}^{N-1} \left(\|a_{\text{req},i|\ell} - a_{\text{req},i|\ell}^0\|_{W_{a_{\text{req}}}}^2 + \|a_{\text{req},i|\ell} - a_{\text{req},i-1|\ell}\|_{W_{\Delta}}^2 \right) \\
& \text{s.t. } z_{i+1|\ell} = Az_{i|\ell} + Ba_{\text{req},i|\ell}, \\
& \quad a_{\text{req},i|\ell} \in \mathcal{A}_r^{\text{tight}}(\hat{x}_{i|\ell}^0), \quad i = 0, \dots, N-1, \\
& \quad z_{i+1|\ell} \in \mathcal{X}, \\
& \quad \|a_{\text{req},i|\ell}\|_{\infty} \leq a_{\text{max}}, \quad \|a_{\text{req},i|\ell} - a_{\text{req},i-1|\ell}\|_{\infty} \leq \dot{a}_{\text{max}}\Delta t.
\end{aligned}$$

Here $\hat{x}_{i|\ell}^0$ is the state on the fixed nominal rollout used to assemble the QP. The state prediction above omits the disturbance term d_{ℓ} of Section III: the implemented propagation is the homogeneous $z_{i+1|\ell} = Az_{i|\ell} + Ba_{\text{req},i|\ell}$, so the predicted position and speed constraints implicitly assume $d_{i|\ell} = 0$ rather than a forecast value. For controllers whose acceleration request does not itself respond to F_h (the PD and AI-conditioned interfaces of Section VI), this makes the state prediction optimistic under load; propagating an explicit disturbance forecast, with a corresponding tightening of the state constraints, is left to future work. The state and acceleration bounds are hard; the implementation contains neither slack variables nor a separately constructed terminal invariant set. The objective trades nominal-command fidelity against acceleration variation, so it may make a small intervention even when the nominal sequence is feasible. Define the first correction as

$$\Delta a_{\text{req},\ell} = a_{\text{req},0|\ell} - a_{\text{req},0|\ell}^0.$$

If the QP solver reports infeasibility, the implementation projects the first nominal acceleration onto the current tightened torque and one-step state polytope and tiles that reactive command over the stored sequence. If this instantaneous polytope is itself empty, it returns zero acceleration. The final high-rate torque projection remains active in either case. This fallback supplies a deterministic bounded command but does not recover horizon feasibility or satisfy the transfer conditions.

The fast loop applies

$$\tau_k = \tau_k^0 + H_r(x_k)\Delta a_{\text{req},k} + \tau_k^{\text{final}},$$

where $H_r(x_k)$ is recomputed from the current state. The slow loop therefore publishes a behavior correction, not a cached full torque.

Let $\hat{\tau}_r(x, a_{\text{req}})$ be the manager's predicted pre-projection torque. Suppose a verified componentwise error bound satisfies

$$\tau_r^{\text{pre}}(x, a_{\text{req}}) \in \hat{\tau}_r(x, a_{\text{req}}) \oplus \mathcal{D}_{\tau,r}(x, a_{\text{req}}),$$

$$\mathcal{D}_{\tau,r}(x, a_{\text{req}}) \subseteq \{\delta\tau : |\delta\tau| \leq \bar{\delta}_{\tau,r}(x, a_{\text{req}})\}.$$

The tightened request set is

$$\mathcal{A}_r^{\text{tight}}(x) = \{a_{\text{req}} : \hat{\tau}_r(x, a_{\text{req}}) \oplus \mathcal{D}_{\tau,r}(x, a_{\text{req}}) \subseteq \mathcal{T}_r\},$$

or, componentwise,

$$\tau_{\min,r} + \bar{\delta}_{\tau,r} \leq \hat{\tau}_r(x, a_{\text{req}}) \leq \tau_{\max,r} - \bar{\delta}_{\tau,r}.$$

The set must include state-estimation error, interpolation, secondary torque, torque-rate limiting, and any other implementation effect that can change the pre-projection torque.

For predicted torque $\tau_{i|\ell}$, define normalized saturation margin, centered so that $\mu = 1$ at τ_c and $\mu = 0$ at either bound,

$$\mu_{i|\ell} = \min_j \left\{ \frac{\tau_{\max,j} - \tau_{j,i|\ell}}{\tau_{\max,j} - \tau_{c,j}}, \frac{\tau_{j,i|\ell} - \tau_{\min,j}}{\tau_{c,j} - \tau_{\min,j}} \right\}, \quad \tau_{c,j} = \frac{\tau_{\max,j} + \tau_{\min,j}}{2},$$

which for symmetric limits reduces to $\mu_{i|\ell} = 1 - \max_j |\tau_{j,i|\ell} - \tau_{c,j}| / \tau_{\max,j}$. This scalar is a diagnostic only and is not computed in the experiments of Section VI; the directional measure below is the one used there.

The minimum horizon margin is

$$\mu_\ell^{\min} = \min_{i=0,\dots,N-1} \mu_{i|\ell}.$$

Scalar utilization does not reveal whether the robot retains authority in the direction required to reject an interaction force. Given current acceleration a_c and unit direction d , define directional authority against the tightened realizable set, not the untightened zonotope $\mathcal{Z}_r(x)$ of Section III, so that the metric accounts for secondary-torque consumption and uncertainty exactly as the optimizer does:

$$\alpha_r^+(x, a_c, d) = \max_{\alpha \geq 0} \{\alpha : a_c + \alpha d \in \mathcal{A}_r^{\text{tight}}(x)\}.$$

A small α_r^+ indicates directional authority collapse even if unused torque remains in other directions.

The implemented manager also reports finite-horizon QP feasibility. This is a useful recoverability diagnostic, but it is not membership in a certified viability kernel because no terminal invariant set or backup policy is constructed. Section VI therefore uses the term *near-boundary braking stress case* rather than point of no return.

V. Conditional Certificate Transfer

Let the abstract behavior model satisfy $z_{\ell+1} = F(z_\ell, a)$ for an action a , and let

$$\mathcal{S} = \{z : V(z) \leq c\}.$$

Rather than fix a single abstract policy, suppose a nonempty *certified action set* is defined at every $z \in \mathcal{S}$,

$$\mathcal{K}_{\text{cert}}(z) = \{a : F(z, a) \oplus \mathcal{E}_\star \subseteq \mathcal{S}\} \neq \emptyset, \quad \forall z \in \mathcal{S},$$

so that $\mathcal{S}$ is robustly invariant under any measurable selection κ with $\kappa(z) \in \mathcal{K}_{\text{cert}}(z)$ for all $z \in \mathcal{S}$. The certificate designer may work with one such selection, but the robot-independent object is the set $\mathcal{K}_{\text{cert}}$, not a distinguished element of it; different robots realizing different members of $\mathcal{K}_{\text{cert}}(z)$ is expected, not a failure of transfer.

For robot r , let $\Pi_r(x) = z$ be the abstraction map and f_r^d its sampled physical dynamics. When no torque projection occurs, suppose the successor mismatch is bounded by

$$\Pi_r(f_r^d(x, \tau_r^{\text{pre}}, F_h)) - F(\Pi_r(x), a_{\text{req}}) \in \mathcal{D}_{z,r}(x, a_{\text{req}}, F_h)$$

for every a_{req} the robot-specific manager may select, not only for one distinguished policy. The set $\mathcal{D}_{z,r}$ may include discretization, interaction-force, state-estimation, interpolation, and secondary-channel errors.

Theorem 1 (Conditional realizability-margin certificate transfer).

Let $\mathcal{K}_{\text{cert}}$ and $\mathcal{S}$ be as above. For robot r , consider an operating region $\mathcal{X}_r$ with $\Pi_r(\mathcal{X}_r) \subseteq \mathcal{S}$. Suppose that for every $x \in \mathcal{X}_r$, every admissible interaction wrench, and every $a_{\text{req}} \in \mathcal{K}_{\text{cert}}(\Pi_r(x))$ that the robot-specific manager may select:

$$\tau_r^{\text{pre}}(x, a_{\text{req}}) \in \hat{\tau}_r(x, a_{\text{req}}) \oplus \mathcal{D}_{\tau,r}(x, a_{\text{req}}), \quad (1)$$

$$\hat{\tau}_r(x, a_{\text{req}}) \oplus \mathcal{D}_{\tau,r}(x, a_{\text{req}}) \subseteq \mathcal{T}_r, \quad (2)$$

$$\mathcal{D}_{z,r}(x, a_{\text{req}}, F_h) \subseteq \mathcal{E}_\star. \quad (3)$$

Assume all implementation effects are included in (1)–(3) and $x_0 \in \mathcal{X}_r$. Then actuator projection is inactive and

$$\Pi_r(f_r^d(x, \tau_r^{\text{app}}, F_h)) \in F(\Pi_r(x), a_{\text{req}}) \oplus \mathcal{E}_\star \subseteq \mathcal{S},$$

the last inclusion following directly from $a_{\text{req}} \in \mathcal{K}_{\text{cert}}(\Pi_r(x))$, with no separate robust-invariance premise about a specific policy required.

Consequently, if the physical successor remains in $\mathcal{X}_r$ for $i = 1, \dots, N$ whenever its abstraction remains in $\mathcal{S}$,

$$\Pi_r(x_{\ell+i}) \in \mathcal{S}, \quad i = 0, \dots, N,$$

which is the finite-horizon guarantee actually supported by the implemented manager and the experiments of Section VI. The unbounded conclusion $\Pi_r(x_k) \in \mathcal{S}$ for all $k \geq 0$ additionally requires the operating-region assumption to hold indefinitely, a separate and independent claim discussed below.

Proof. Conditions (1)--(2) imply $\tau_r^{\text{pre}} \in \mathcal{T}_r$; hence projection is the identity and $\tau_r^{\text{app}} = \tau_r^{\text{pre}}$. Condition (3) places the concrete successor inside $F(\Pi_r(x), a_{\text{req}}) \oplus \mathcal{E}_\star$, which lies in $\mathcal{S}$ directly by the definition of $\mathcal{K}_{\text{cert}}$ since $a_{\text{req}} \in \mathcal{K}_{\text{cert}}(\Pi_r(x))$. Finite induction from $x_0 \in \mathcal{X}_r$ over $i = 0, \dots, N$ gives the stated horizon result; extending it to all $k \geq 0$ requires the operating-region assumption to hold at every step, which this argument alone does not establish. $\square$

Remark (connection to the implemented manager). The QP of Section IV enforces $a_{\text{req},i|\ell} \in \mathcal{A}_r^{\text{tight}}(\hat{x}_{i|\ell}^0)$; it does not enforce $a_{\text{req},i|\ell} \in \mathcal{K}_{\text{cert}}(z_{i|\ell})$, and the two sets are constructed independently rather than intersected inside the solver. Theorem 1 therefore certifies the manager's actual output only on samples where that output happens to also lie in $\mathcal{K}_{\text{cert}}$, which the present experiments do not verify online; the audit of Section VI checks the realizability conditions (1)--(2) and the sampled successor defect, not $\mathcal{K}_{\text{cert}}$ membership. Constructing a concrete, computable representation of $\mathcal{K}_{\text{cert}}$ and adding it as an explicit QP constraint would close this gap between the theorem and the algorithm; this reduced-order study does neither and leaves both to future work.

The region-persistence clause is likewise an explicit conditional assumption, not a consequence of the implemented finite-horizon QP. A recursively feasible terminal set or certified backup policy would be one way to establish it over an unbounded horizon. Without such a construction, the horizon in this study should be read as finite --- matching the manager's own $N = 12$ -step horizon --- rather than indefinite.

The theorem does not make feasibility universal. Each robot must verify $\Pi_r, \hat{\tau}_r, \mathcal{D}_{\tau,r}, \mathcal{D}_{z,r}$, and $\mathcal{X}_r$. What transfers is the abstract object

$$(F, \mathcal{K}_{\text{cert}}, \mathcal{S}, \mathcal{E}_\star).$$

A directly testable norm specialization is

$$\bar{\eta}_{\text{disc}} + \bar{\eta}_{\text{hold}} + \bar{\eta}_{\text{sec}} + L_F \bar{\delta}_F + L_\tau \bar{\delta}_\tau \leq \bar{\epsilon}_\star,$$

where L_F bounds the one-step abstraction defect induced by force-model error and L_τ bounds the defect induced by torque-realization error in the selected norm, together with

$$\min_j \{ \hat{\tau}_j - \tau_{\min,j}, \tau_{\max,j} - \hat{\tau}_j \} \geq \|\bar{\delta}_\tau\|_\infty.$$

The first inequality requires the certificate radius to dominate the successor defect; the second requires actuator margin to dominate torque-realization uncertainty.

If clipping becomes active, let

$$\mathcal{C}_{\tau,r}(x, a_{\text{req}}) = \{ \text{proj}_{\mathcal{T}_r}(\tau) - \tau : \tau \in \hat{\tau}_r(x, a_{\text{req}}) \oplus \mathcal{D}_{\tau,r}(x, a_{\text{req}}) \}.$$

Here $\mathcal{L}_r(x)$ is a verified linear or set-valued one-sample sensitivity operator that maps a torque-projection residual into the resulting abstraction-space successor perturbation. Its induced norm is bounded by the scalar gain L_τ used above.

Transfer can then hold only if the enlarged defect

$$\mathcal{D}_{z,r} \oplus \mathcal{L}_r(x) \mathcal{C}_{\tau,r} \subseteq \mathcal{E}_\star.$$

The experiments use the more transparent no-clipping branch of Theorem 1. When the request leaves the tightened set, the final projection protects the actuator but no longer implies preservation of the requested behavior certificate.

A further direction, sketched here but not developed as a formal result, is task-channel dissipativity transfer: if the abstract certificate additionally supplied a storage function on the task-force channel, and the physical realization were shown power-consistent with it up to a bounded defect on the no-clipping branch, the resulting dissipation budget would transfer alongside the state certificate above. Realizing this rigorously requires relating the abstract storage function to a physical robot storage function, fixing sign conventions between the task-error convention e and physical interaction power, and bounding the clipping-branch saturation-power term via the directional-authority margin α_r^+ ; none of this is constructed, related to a physical storage function, or evaluated in the present study, and we leave it to future work rather than state it as a corollary.

VI. Experiments

A. Protocol

The deterministic benchmark uses a two-dimensional interaction task. The fast controller and final projection execute at 1 kHz; the manager executes at 50 Hz with horizon $N = 12$. Three

configuration-dependent realization maps are evaluated: a planar 2R map, an FR3-inspired surrogate, and a six-axis-arm surrogate. The latter two reproduce different actuator geometries and limits but are not manufacturer-accurate rigid-body models.

The same behavior dynamics, predictive objective, and sampled audit threshold

$$\epsilon_{\text{audit}} = 0.03 \text{ m/s}$$

are used throughout. This scalar is an empirical acceptance threshold for the observed norm $\|d_z\|_2$; it is deliberately not denoted by the certified set $\mathcal{E}_\star$ of Section V. The simulation does not construct V , $\mathcal{S}$, or a workspace-wide proof of robust invariance. The fast-controller interfaces are PD, impedance, a small policy trained by deterministic evolution strategy, a fixed-feature neural policy fitted to impedance demonstrations, and an AI-conditioned motion primitive executed by a PD servo. The final two learned cases test the command/preview interface; they are not evidence of semantic AI safety or improved policy quality.

The 108 deterministic configurations comprise 40 scenario cases, 30 controller-interface cases, 24 cross-realization cases, and 14 ablations. The 40 scenario cases are eight scenarios evaluated with five channels: four realization architectures—direct clipping, a reactive 1 kHz projection, a scalar reference governor followed by the same reactive projection, and the proposed horizon-wide correction with final actuator projection—plus one nominal diagnostic without torque projection. The stress scenarios are no saturation, slow saturation, sudden disturbance, directional authority collapse, near-boundary braking, model mismatch, preview mismatch, and a dedicated horizon-ramp scenario used only for the horizon-constraint ablation of Section VI.E.

We report pre-projection torque excess, applied torque excess, workspace excess, behavior-realization RMSE, warning lead time, directional authority, sampled successor defects, and computation time. For a two-dimensional residual r_k , RMSE is the pooled component-wise quantity

$$\text{RMSE}_{\text{comp}} = \sqrt{\frac{1}{2K} \sum_{k=1}^K \|r_k\|_2^2}.$$

The successor defect in Table III is instead the maximum vector norm and is not directly comparable with the component-wise RMSE. Applied torque remains within its box whenever the final projection is active; zero pre-projection excess is one sampled check for the no-clipping branch of Theorem 1, not proof of the abstract robust-invariance premise.

B. Anticipatory saturation management

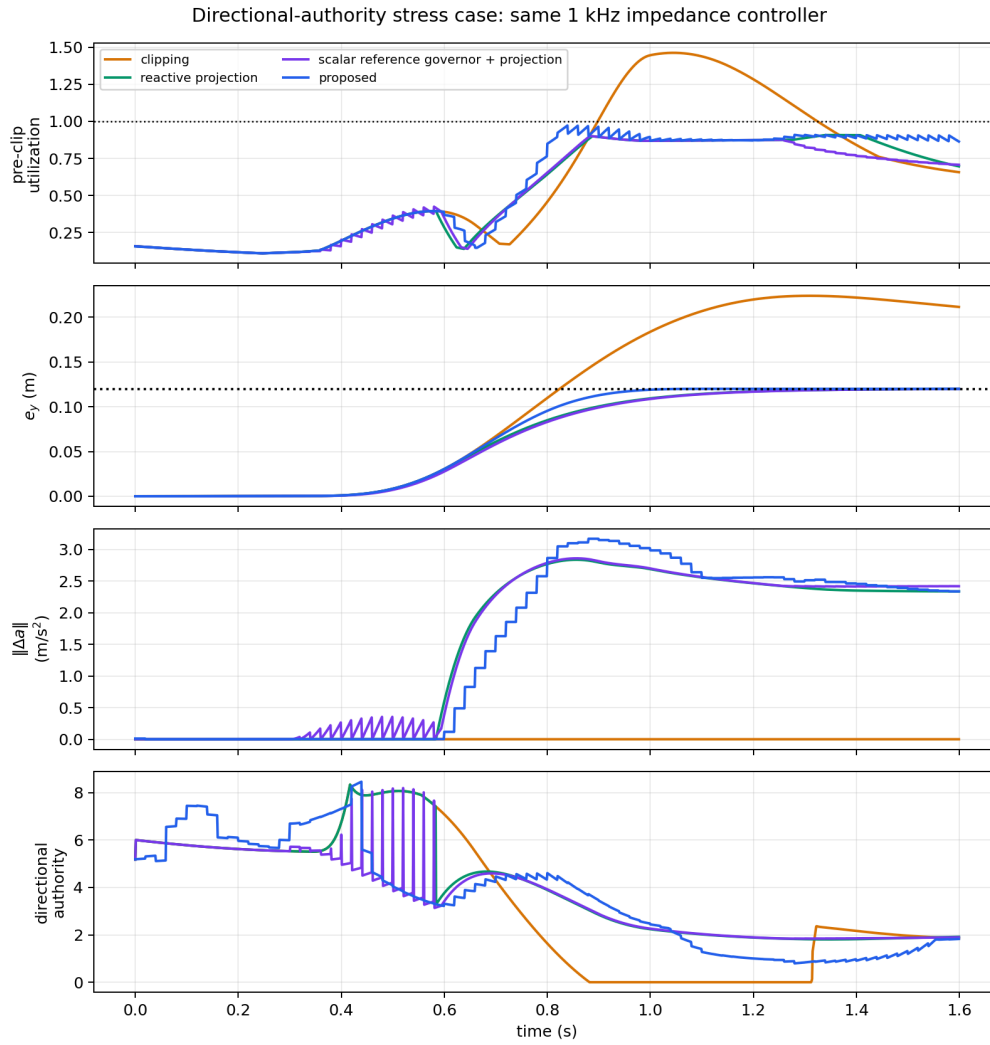


Fig. 1 shows the directional-authority stress case. Direct clipping exceeds the actuator-feasible request set and the workspace bound. The predictive methods intervene earlier, but the vector correction preserves the workspace constraint while retaining a visible intervention residual.

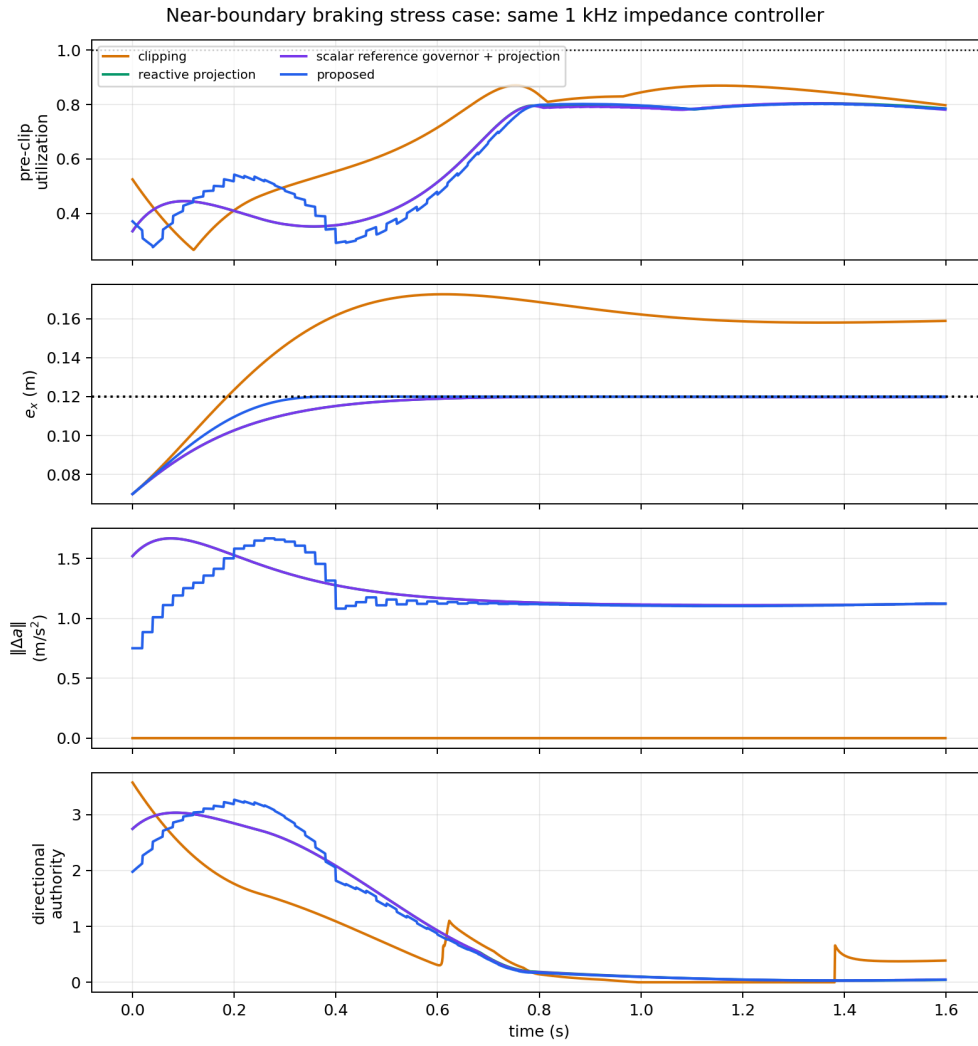


Fig. 2 shows the near-boundary braking case, which starts with an outward velocity close to the position boundary under a shrinking torque budget. Direct clipping lets the position overshoot the boundary and settle outside it. The reactive projection, scalar reference governor plus projection, and proposed manager all arrest the position at the boundary. Because the experiment does not compute a viability kernel or terminal invariant set, it supports only finite-horizon constraint handling in this stress case.

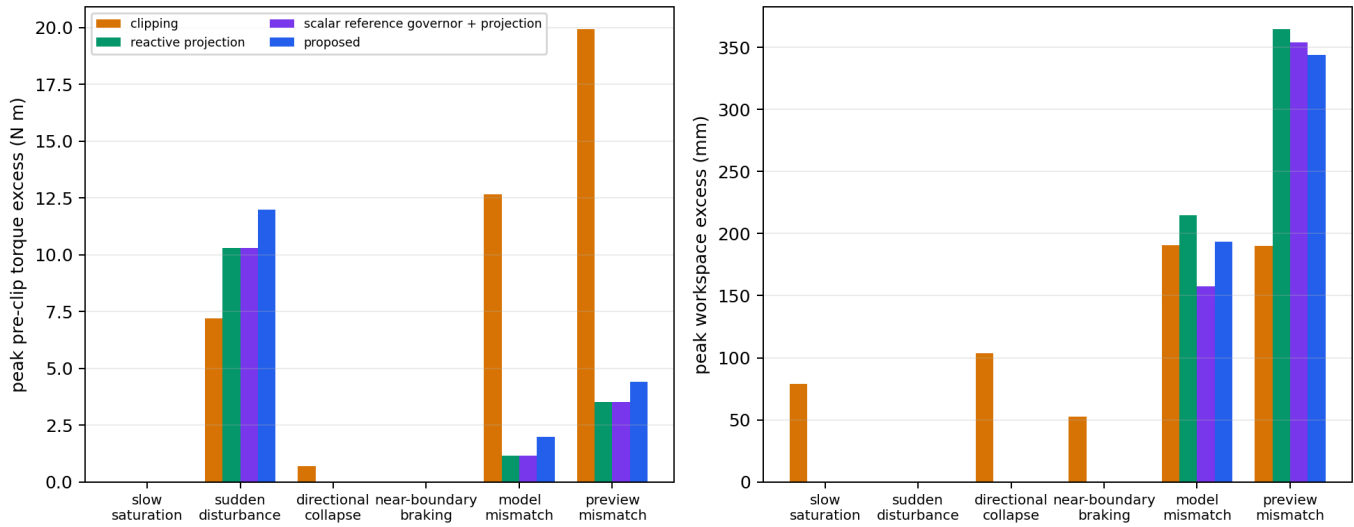


Fig. 3 summarizes the method-level trends, while Table I reports the clipping and proposed results. The proposed manager preserves the no-saturation behavior and prevents workspace violations in the slow-saturation, directional-collapse, and near-boundary braking scenarios. Warning precedes the limiting event by 0.912, 0.939, and 0.566 s, respectively. The reactive projection and scalar reference governor plus projection also satisfy the sampled constraints in these three cases. For slow saturation their warning leads are 0.419 and 0.395 s, and for directional collapse they are 0.340 and 0.306 s. These lead-time differences are descriptive results against the implemented scalar governor, not evidence of superiority over directional or vector reference governors. In the near-boundary braking case, all three methods warn at approximately 0.57 s.

Scenario	Pre-proj. C/P (Nm)	Workspace C/P (mm)	Lead (s)	QP feasible	Audit
No saturation	0.000 / 0.000	0.000 / 0.000	--	100%	Yes
Slow saturation	0.000 / 0.000	78.970 / 0.001	0.912	100%	Yes
Sudden disturbance	7.208 / 11.976	0.000 / 0.000	0.084	92.5%	No
Directional collapse	0.693 / 0.000	103.914 / 0.003	0.939	100%	Yes
Near-boundary braking	0.000 / 0.000	52.537 / 0.011	0.566	100%	Yes
Model mismatch	12.665 / 1.981	190.434 / 193.599	0.558	45.0%	No
Preview mismatch	19.917 / 4.409	190.191 / 344.246	0.790	40.0%	No

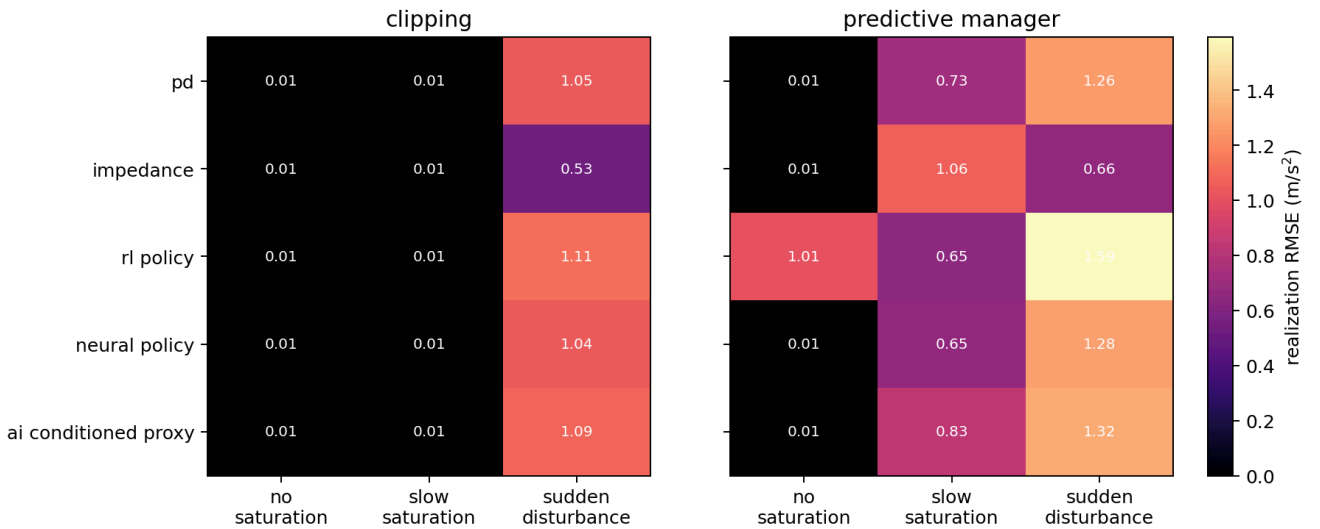
: Scenario results.

In Table I, C/P denotes clipping/proposed, QP feasibility is the fraction of 50 Hz updates whose horizon problem is feasible, and Audit denotes whether all sampled realization checks pass. The QP is feasible at every update in the four successful stress cases, but only 74/80, 36/80, and 32/80 updates in sudden disturbance, model mismatch, and preview mismatch, respectively. Infeasible updates use the reactive fallback defined in Section IV, so the reported “proposed” trajectory includes that fallback and must not be interpreted as horizon-MPC behavior throughout.

The negative cases are not merely inconclusive. Under preview mismatch, the proposed correction acts on a force forecast that misses the sign change inside the horizon, increasing workspace excess from 190.191 mm with clipping to 344.246 mm. Under model mismatch it is also slightly worse, 193.599 versus 190.434 mm. The sampled interface audit rejects both cases, correctly indicating that the anticipatory correction is not trustworthy there.

The sudden-disturbance result illustrates why the slow manager cannot be the only protection layer. The wrench changes without advance information, while the implemented preview holds the measured wrench constant over the horizon; the resulting correction can therefore be misaligned with the short impulse and raises pre-projection excess from 7.208 to 11.976 Nm. The final projection keeps the applied actuator command inside its box, but the pre-projection request is infeasible and the observed successor defect exceeds ϵ_{audit} . These results identify operating conditions for which Theorem 1 cannot be invoked.

C. Controller-interface substitution



As illustrated in Fig. 4, the manager formulation and weights are unchanged across controller interfaces. Under no saturation, correction RMSE is below 0.01 m/s^2 for four of the five controllers. The small evolution-strategy policy is the exception, with correction RMSE $\approx 1.01 \text{ m/s}^2$. Its limited

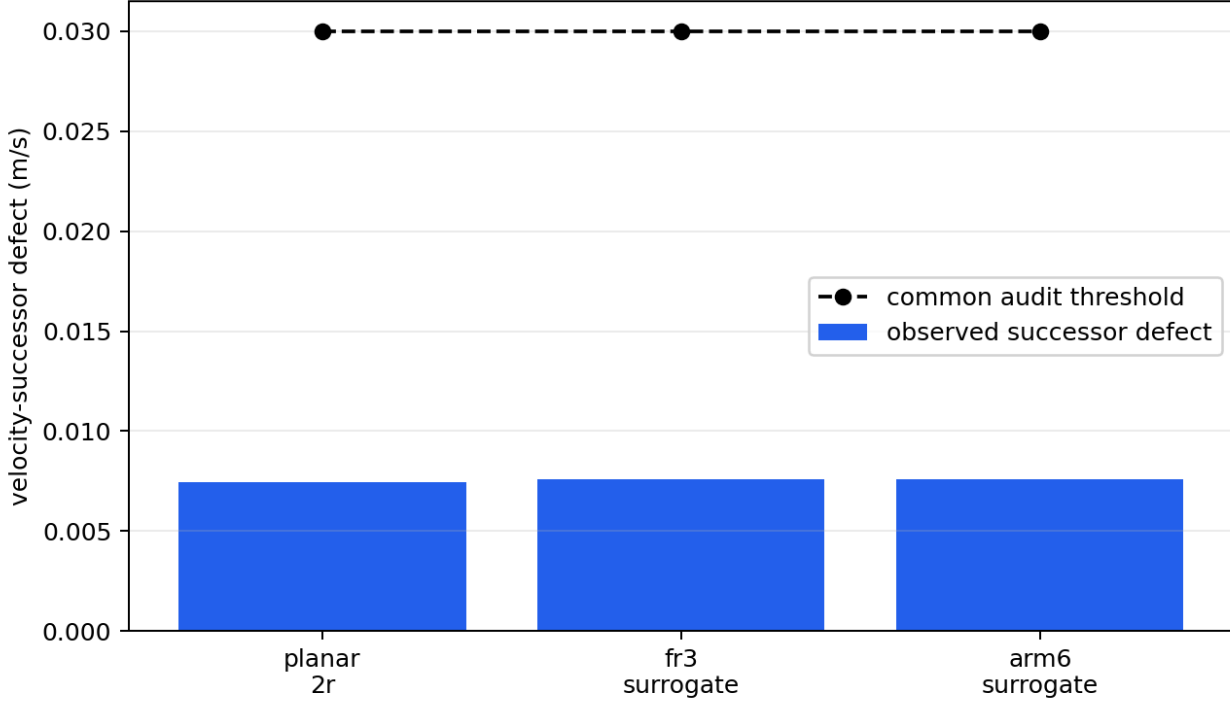
training set does not cover the test trajectory symmetrically, and the learned policy retains a positive- y command bias. This is an observed generalization error rather than evidence of a specific failure of the evolution-strategy algorithm. The manager holds the state inside the workspace box, but is compensating for a biased interface rather than remaining inactive. This case is retained as an interface stress test: the architecture is not intended to repair behavior-policy quality, although its running constraints may incidentally reject a biased request.

Under slow saturation, realization RMSE is nearly equal to correction RMSE for every proposed-controller case (for example, 0.732 versus 0.726 m/s² for PD). The reported residual is therefore almost entirely the manager's deliberate intervention, not unmodeled tracking error. Clipping appears to have a smaller realization RMSE because it follows the nominal request instead of enforcing the workspace constraint; in the impedance case that apparent fidelity accompanies 78.970 mm of workspace violation, whereas the proposed manager leaves 0.001 mm. All five proposed cases pass the sampled interface audit (Table II). The results demonstrate substitution at the command/preview interface; they do not certify an arbitrary learned policy whose preview error is unbounded.

Interface	Realization RMSE (m/s ²)	Correction RMSE (m/s ²)	Lead (s)	Excess (mm)
PD	0.732	0.726	0.991	0.001
Impedance	1.064	1.060	0.912	0.001
Trained policy	0.652	0.652	1.207	0.016
Fitted neural policy	0.655	0.649	1.016	0.001
AI-conditioned proxy	0.835	0.830	0.959	0.006

: Controller substitution under slow saturation.

D. Sampled interface audit across realization maps



The behavior model, predictive objective, and 0.03 m/s audit threshold remain unchanged across the three realization maps. Only $\hat{\tau}_r$, the actuator box, and the observed torque- and successor-error checks change. As shown in Fig. 5 and Table III, every observed successor defect is below 0.00759 m/s , leaving more than 0.0224 m/s of the audit allowance unused.

Realization map	Max. observed (m/s)	Unused audit (m/s)	Min. bound slack (Nm)
Planar 2R	0.007456	0.022544	0.002525
FR3-inspired surrogate	0.007586	0.022414	8.82×10^{-6}
Six-axis-arm surrogate	0.007586	0.022414	9.97×10^{-5}

: Sampled cross-realization interface audit.

The last column is not remaining actuator authority. It is the smallest sampled containment residual $\bar{\delta}_{\tau,j} - |\tau_j^{\text{pre}} - \hat{\tau}_j|$. For the cross-realization cases, the componentwise bounds use

$$\bar{\delta}_{\tau} = 0.03 \mathbf{1}_{n_r} + 0.008 \max(|H_r(0,0)|, 0.25) \left| a_{\text{req}} - \frac{F_h}{m_r} \right|,$$

where the maximum is elementwise. These bounds range from 0.0300 to 0.0917 N $\cdot$ m over the sampled audit. The near-zero FR3 and six-axis residuals mean that the deterministic injected errors nearly attain their envelopes; they do not mean that the tightening is zero. The minimum planned actuator margins are 0.233, 0.679, and 0.741 N $\cdot$ m for the planar, FR3-inspired, and six-axis maps, respectively. Thus error-bound containment is the tight numerical check, while actuator authority is not binding on these sampled trajectories. Because the injected errors are constructed from the same envelopes, this is a consistency audit rather than independent validation of $\mathcal{D}_{\tau,r}$.

This is a sampled-trajectory interface audit, not independent uncertainty validation, an analytic whole-workspace proof, or an experimental proof of robust invariance. It demonstrates only that the same numerical audit threshold and checking mechanism can be applied to multiple robot-specific realization maps. A full certificate-transfer experiment would additionally require independently identified uncertainty sets and a verified $(F, \mathcal{K}_{\text{cert}}, \mathcal{S}, \mathcal{E}_\star)$.

E. Ablations and computation

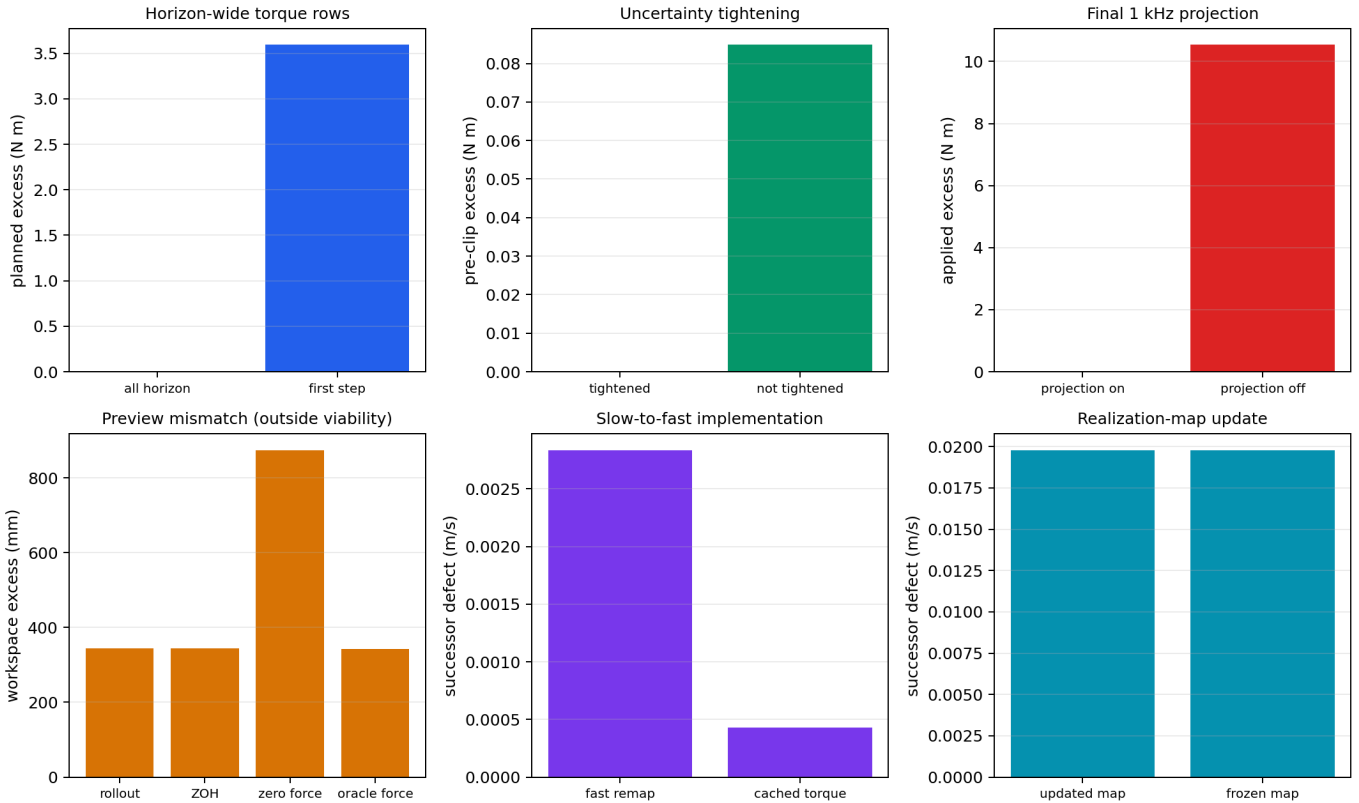


Fig. 6 summarizes the ablations. Constraining all predicted moves eliminates planned torque excess in the horizon-ramp case. Constraining only the first move preserves the currently applied command but leaves a 3.594 N $\cdot$ m future violation and produces 31.192 mm workspace excess. This result directly verifies the predictive-horizon constraint requirement.

Removing uncertainty tightening creates 0.0848 Nm pre-projection excess. The final projection hides that excess at the applied actuator channel but does not restore the no-clipping refinement condition. Conversely, disabling the final projection under sudden disturbance produces 10.538 Nm applied excess, confirming that prediction and high-rate protection have distinct responsibilities.

The preview and implementation ablations also identify non-results. Zero-force preview increases workspace excess from 344.25 to 874.06 mm, but no preview option restores viability in the severe mismatch case. In this reduced-order model, recomputing the fast realization map does not outperform cached torque, and updating the realization map is numerically indistinguishable from freezing it. Benefits from these mechanisms therefore remain to be demonstrated on nonlinear rigid-body systems.

Increasing the manager rate from 50 to 100 Hz could shorten the interval over which an obsolete correction is held, but it would not predict an unannounced wrench change or repair an incorrect force model. Addressing the sudden-disturbance limitation therefore requires both timing and information improvements—for example, event-triggered re-solving, bounded disturbance observers, or preview uncertainty propagated into the horizon—while retaining the 1 kHz final projection.

In the final regenerated run, the manager's median-of-run-medians is 1.795 ms, and its worst observed maximum is 13.543 ms, below its 20 ms period. The fast path has a median-of-run-medians of 121.7 μ s and a worst observed maximum of 451.7 μ s. All recorded calls met their nominal periods on this run, but non-real-time Python measurements establish typical throughput, not hard real-time execution.

VII. Discussion

The predictive optimization statement and empirical audit threshold remain unchanged across the implemented controller interfaces and realization maps. Physical feasibility does not transfer automatically: each robot must reconstruct and verify its actuator-feasible set and its refinement-error bounds.

This distinction also clarifies the relation to a reference governor or predictive safety filter. Those architectures can adopt the same behavior coordinates. What separation adds is an explicit proof boundary: a reusable behavior certificate on one side and a checkable robot-specific realization contract on the other. In the present study only an interface mechanism is sampled; the abstract certificate and independent uncertainty identification are absent. If the torque uncertainty exceeds the

available margin, or an observed successor defect exceeds ϵ_{audit} , the experimental audit rejects the case. This rejection is evidence of a failed sampled check, not proof about the theoretical set $\mathcal{E}_*$.

The final high-rate projection is essential but should not be confused with behavior preservation. It enforces the actuator box after an unexpected disturbance, while Theorem 1 states when projection remains inactive and the requested closed-loop behavior is actually realized. The mismatch experiments show that these two properties can diverge.

For model-based physical AI, the architecture provides a runtime boundary between behavior generation and physical realization. Learned, diffusion-based, or language-conditioned modules may propose behavior, while the realization model evaluates what the current robot can execute. This contract does not certify the semantics or intent of an AI-generated command; it certifies only the modeled physical refinement inside the verified operating region.

Several limitations remain. First, the robot substitutions are reduced-order actuator-geometry surrogates rather than full rigid-body or hardware systems. Second, the reported errors are observations along experiment trajectories rather than certified bounds over a continuous workspace. Third, the benchmark does not instantiate the robust-invariance premise of Theorem 1; “audit threshold” and “observed defect” are therefore kept distinct from the theoretical certificate set. Fourth, the benchmark omits orientation, redundant null-space tasks, sensor delay, contact transitions, state-estimation uncertainty, human-participant validation, and any construction or evaluation of the task-channel dissipativity sketch of Section V. Fifth, the reference-governor baseline includes a reactive projection and is an architecture-level comparator, not a reproduction of every established governor design. Its scalar command parameterization is especially restrictive in the directional-collapse case; a directional or vector reference governor would be expected to narrow the reported lead-time difference. Sixth, every reported case is a single deterministic trajectory per scenario, controller, and realization map; the study does not sweep disturbance magnitude, timing, or sensor noise to characterize how close a successful case is to its failure boundary, so the reported margins are point estimates rather than statistically characterized safety margins. Finally, the theorem's region-persistence clause and recursive feasibility are the same unresolved gap at different levels: discharging that assumption and defining a certified point of no return require a formally constructed terminal invariant set or backup policy. Finite-horizon running constraints alone do not imply global recoverability.

VIII. Conclusion

This paper introduced a predictive realization architecture for actuator-limited fast robot controllers. The nominal controller remains at 1 kHz, while a slower MPC layer forecasts robot-specific loss of realizability and modifies the requested acceleration before clipping. The proposed separation does not make actuator feasibility universal. It identifies which theoretical object may be reused—an independently established abstract behavior certificate—and which objects must be verified again—the realization map, actuator margin, uncertainty bounds, and operating region.

Across 108 deterministic reduced-order cases, full-horizon correction detects a future violation missed by a first-step constraint and accepts five nominal-controller interfaces. A sampled interface audit applies the same successor-defect threshold across three realization maps. Slow saturation, directional authority collapse, and near-boundary braking violations are prevented within the tested region. Abrupt disturbance and severe mismatch expose cases where the sampled audit fails, the QP frequently becomes infeasible, and the reactive fallback plus final actuator projection remain available.

Future work will replace the surrogate maps with full rigid-body systems, certify uncertainty sets over continuous workspaces, construct terminal invariant sets, and evaluate the architecture in a real-time hardware loop. Within the present model, three gaps remain open: the QP does not yet propagate the estimated disturbance $d_{i|\ell}$ of Section III through its state prediction, so predicted-state constraints implicitly assume it is zero; the manager's feasible set $\mathcal{A}_r^{\text{tight}}$ is not intersected with the certified action set $\mathcal{K}_{\text{cert}}$ inside the solver, so Theorem 1 is not yet enforced by construction; and the task-channel dissipativity sketch of Section V requires a constructed physical storage function and a directional-authority bound on the clipping-branch saturation-power term before it can be stated as a corollary. These steps, together with hardware validation, are necessary before claiming hardware-level certificate transfer or interaction safety.

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